

LoRA Rank and Domain Adaptation of Stable Diffusion 1.5 to Cityscapes

Progress report

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1 Objective

Measure how LoRA rank affects domain adaptation of Stable Diffusion 1.5 to the Cityscapes visual domain, evaluated with FID, KID, CLIPScore, BLIP image-text matching and LPIPS diversity against real Cityscapes imagery.

Ranks under study: $\{8, 16, 32, 64, 128\}$, plus a frozen SD 1.5 baseline and the real images as reference.

2 Dataset and preprocessing

Split	Source images	Cities	Cropped
<code>train</code>	2 975	18	2 975
<code>val</code>	500	3	500
<code>test</code>	1 525	6	1 525
<code>train_extra</code>	19 997	23	19 995
Total	24 997	50	24 995

Table 1: Cityscapes leftImg8bit. Two images were rejected (see below).

Cropping. Every source frame is 2048×1024 . We take the central 1024×1024 square with PIL crop box $(512, 0, 1536, 1024)$ — a crop, not a resize, so no aspect distortion. All 24 995 outputs were verified to be exactly 1024×1024 .

Rejections. A validity guard skipped zero-byte files, undecodable files, and frames whose mean pixel value was < 5 (all-black). Beyond the one known corrupt frame (`troisdorf_000000_000073`, deleted previously), the guard caught **two further all-black frames**: `heidelberg_000000_000293` (mean 4.97) and `oberhausen_000000_000899` (mean 4.58). Hence $24\,997 - 2 = 24\,995$.

Data split policy. We do *not* hold out a test split. Section 4 explains why this does not risk memorisation, and why holding out would cost more in metric precision than it buys in purity.

3 Captioning

Florence-2-large with task `<DETAILED_CAPTION>`; 3 beams; `max_new_tokens = 1024`; greedy decoding. Leading boilerplate (“The image shows/depicts/features”, “This image shows”) was stripped and the first letter re-capitalised.

Captions generated	24 995
Distinct captions	18 595 (74.4%)
Captions used more than once	1 819, covering 8 219 images
Largest duplicate group	105 images share one caption
Length (words)	mean 35.0, median 35, range 14–75
Length (CLIP tokens)	mean 43.3, median 43, p99 66, max 92
Exceeding CLIP’s 77-token limit	15 images (0.06%)

Table 2: Caption statistics. Runtime 3 h 52 m on one A100.

4 Experimental design: prompts, seeds and pairing

This section answers the question of what is held constant across conditions.

Index convention. The caption file `metadata.jsonl` is ordered `test` $\rightarrow$ `train` $\rightarrow$ `train_extra` $\rightarrow$ `val`, with cities and filenames sorted. Its *line number is the image index*:

<code>test</code>	indices 0–1 524	(1 525)
<code>train</code>	indices 1 525–4 499	(2 975)
<code>train_extra</code>	indices 4 500–24 494	(19 995)
<code>val</code>	indices 24 495–24 994	(500)

Are the prompts the same across ranks and checkpoints? Yes — exactly. Image i in *every* generated set is produced from caption line i with generator seed i :

```
generator = torch.Generator("cuda").manual_seed(i).
```

Consequently prompt 4217 receives the identical caption *and* the identical initial latent noise under the frozen model, under every rank, and under every checkpoint. The only variable that differs between conditions is the adapter. This makes the comparison **paired**: per-image metrics (CLIP, BLIP) admit a paired test, and set-level metrics (FID, KID) are not contaminated by prompt-sampling variance.

We verified on disk that for index 4714 the frozen and rank-8 mappings agree with caption line 4714, and that all 50 checkpoint directories contain an *identical* index set.

Checkpoint-sweep sample. Scoring all 24 995 prompts for every rank $\times$ checkpoint cell is infeasible, so each cell uses a strided subsample, `stride` = 50: indices 0, 50, 100, ..., 24 950, i.e. **500 images per cell**. A contiguous head would have been all Berlin test images; the stride instead reproduces the corpus composition almost exactly:

Split	In 500-sample	Full corpus
<code>test</code>	31 (6.2%)	6.1%
<code>train</code>	59 (11.8%)	11.9%
<code>train_extra</code>	400 (80.0%)	80.0%
<code>val</code>	10 (2.0%)	2.0%

50 cities are represented. The same 500 indices are used in all 50 cells and for the frozen baseline, so every number in Section 7 is computed on identical inputs.

Why no held-out split. Caption duplication makes memorisation-via-caption structurally implausible: a caption does not identify an image (one caption maps to up to 105 distinct frames), so conditioning on a training caption specifies a broad distribution rather than retrieving a specific

image. FID/KID are distributional, not per-image recall. Cityscapes splits are near-identically distributed (same capture campaign and camera rig), so a held-out set would shift the numbers little while reducing n to 2025, where FID is badly biased.

5 Training

HuggingFace `train_text_to_image_lora.py` (diffusers 0.36.0), vendored and patched to cast LoRA parameters to fp32 after `unet.add_adapter`. Identical hyperparameters for all five ranks:

base model	<code>stable-diffusion-v1-5</code>	LR schedule	cosine
resolution	768	peak learning rate	2×10^{-4}
batch (per device)	2	<code>snr_gamma</code>	5.0
GPUs	2 (A100)	<code>max_grad_norm</code>	1.0
grad. accumulation	2	seed	42
effective batch	8	<code>max_train_steps</code>	31 000
precision	fp16	<code>lora_alpha</code>	= rank (scale 1)
target modules	<code>to_k, to_q, to_v, to_out.0</code> — 128 modules		

31 000 steps over 24 995 images at effective batch 8 is ≈ 10 epochs.

Learning-rate schedule. The learning rate is *not* constant during training: it warms up linearly over 1 200 steps to a peak of 2×10^{-4} , then follows a cosine decay to approximately zero at step 31 000. Values recovered from the rank-8 training log (31 001 samples):

step	1	600	1 200	5 000	10 000	15 000
LR	1.67×10^{-7}	1.00×10^{-4}	2.00×10^{-4}	1.92×10^{-4}	1.60×10^{-4}	1.12×10^{-4}
step	20 000	25 000	29 000	30 999		
LR	6.00×10^{-5}	1.94×10^{-5}	2.23×10^{-6}	2.72×10^{-11}		

The peak value 2×10^{-4} is *identical for all five ranks*; only the schedule’s peak, not its shape, is at issue in the caveat raised in Section 9.

Rank	Wall time	Adapter size	Parameters	Checkpoints	$\sum \ BA\ _F$
8	4:38:12	6.1 MB	1 594 368	10	867.8
16	4:38:55	12.2 MB	3 188 736	10	719.4
32	4:38:37	24.4 MB	6 377 472	10	688.1
64	4:40:28	48.7 MB	12 754 944	10	771.2
128	4:43:26	97.3 MB	25 509 888	10	935.2

Table 3: All five ranks trained successfully. Wall time is nearly constant across a $16\times$ range of rank, confirming the adapter is a negligible share of compute next to the frozen UNet. $\sum \|BA\|_F$ is the total effective weight update applied to the UNet; `lora_alpha=rank` means the scale factor is 1, so these are directly comparable.

Checkpoints were saved every 3 000 steps, giving 10 per rank (3 000–30 000).

6 Generation

All images: 768×768 , 30 inference steps, guidance scale 7.5 (default), fp16, safety checker disabled, `seed` = prompt index.

Set	Images	Status
Frozen SD 1.5 (baseline)	24 995 / 24 995	complete
rank 16, final weights	24 995 / 24 995	complete
rank 32, final weights	24 995 / 24 995	complete
rank 8, final weights	15 918 / 24 995	partial
rank 64, final weights	13 647 / 24 995	partial
rank 128, final weights	0 / 24 995	not started
Checkpoint sweep	50 dirs $\times$ 500 = 25 000	complete

Table 4: Generated image sets. The final-weight sets were halted once Section 8’s over-training finding showed the final checkpoint is the worst one.

7 Evaluation

Method. Every set is scored on the *intersection* of available indices across the sets being compared, so n and the prompt list are identical everywhere — necessary because FID is strongly sample-size biased. Real reference crops are resized once from 1024^2 to 768^2 so real and generated images enter the metric from the same pixel grid.

Metrics: FID and KID via `clean-fid (mode="clean")`; CLIPScore as $100 \cdot \max(0, \cos(E_I, E_T))$ from `clip-vit-large-patch14`; BLIP ITM probability and cosine from `blip-itm-base-coco`; LPIPS (AlexNet) mean pairwise distance within a set as a diversity proxy.

Noise floor. Splitting the real images in half and scoring them against each other gives the lowest achievable value at this sample size: **FID 66.17**, **KID 0.00115** ($n = 250$ per half). FID cannot approach 0 at small n ; KID, being unbiased, does.

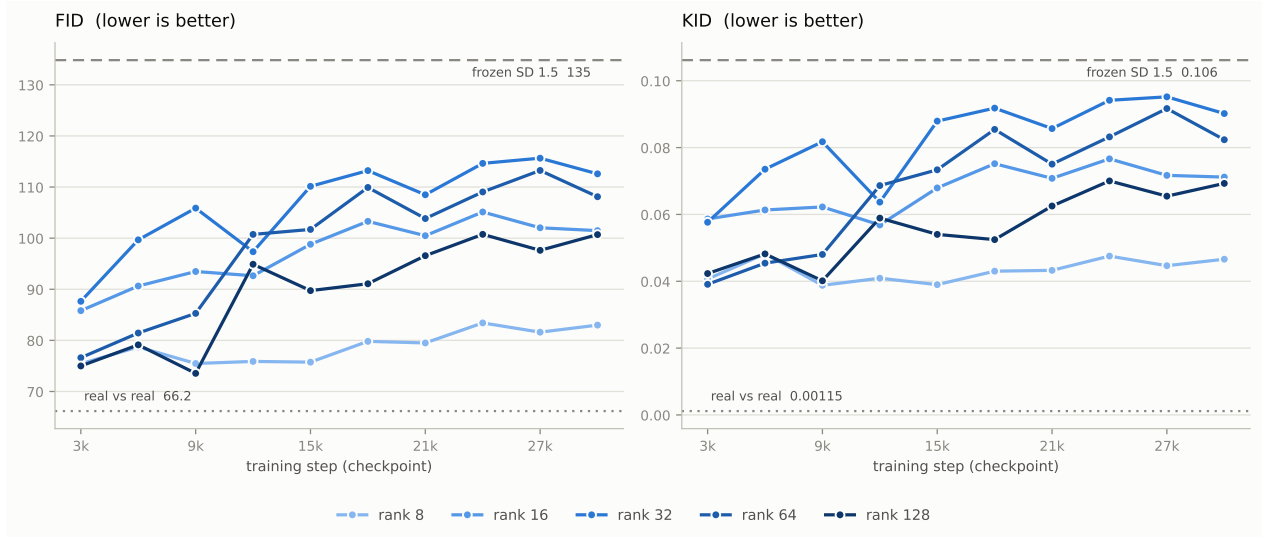


Figure 1: FID and KID against training checkpoint, one line per rank ($n = 500$ identical prompts and seeds per point). Dashed line: frozen SD 1.5. Dotted line: real-vs-real noise floor.

8 Results

Three findings.

- LoRA adaptation works, and strongly.** KID falls from 0.1062 (frozen) to ≈ 0.039 – 0.058 , a 2 – $3\times$ improvement. FID falls from 134.8 to ≈ 73 – 88 against a floor of 66.2 — i.e. the best

Rank	Best by FID		Best by KID	
	Checkpoint	FID	Checkpoint	KID
8	3 000	75.43	9 000	0.0388
16	3 000	85.81	12 000	0.0569
32	3 000	87.64	3 000	0.0577
64	3 000	76.61	3 000	0.0391
128	9 000	73.55	9 000	0.0401
frozen SD 1.5	—	134.83	—	0.1062
real vs real	—	66.17	—	0.00115

Table 5: Best checkpoint per rank, from the 50-cell sweep.

adapted models close most of the gap to real images.

2. **The optimum is early; 31 000 steps is far too many.** Every rank peaks at checkpoint 3 000 or 9 000 and degrades monotonically thereafter. Training ran ≈ 10 epochs when $\approx 1\text{--}3$ would have sufficed. This is visible in the images: at step 3 000 the Mercedes hood ornament (present in every real Cityscapes frame, since the dataset was captured from a Mercedes) is rendered sharply; by step 30 000 it degenerates into a cyan blob, with magenta patches on road surfaces. Degradation is worse at higher rank, consistent with `lora_alpha=rank` (scale $\alpha/r = 1$) meaning the same peak LR produces larger effective updates as rank grows. Note the cosine decay puts the last $\approx 6\,000$ steps below 2×10^{-6} , so the degradation accumulates during the high-LR middle of the run (roughly steps 3 000–20 000), not at the tail.
3. **There is no monotonic rank trend.** Ranks 8, 64 and 128 perform similarly (FID $\approx 73\text{--}77$); ranks 16 and 32 are notably worse ($\approx 86\text{--}88$). Capacity is evidently not the binding constraint — supported independently by $\sum \|BA\|_F$, which *falls* from rank 8 (867.8) to rank 32 (688.1) before rising again: doubling capacity did not produce a larger functional change.

9 Current status and next steps

Complete: preprocessing; captioning; frozen baseline (24 995); all five adapters trained with 10 checkpoints each; the 50-cell checkpoint sweep (25 000 images) and its FID/KID evaluation.

Running: full metric suite (FID, KID, CLIP, BLIP, LPIPS) on frozen plus all five ranks at checkpoints 3 000 and 9 000.

Next:

1. Fix the evaluation checkpoint (3 000 or 9 000) from the sweep and the full-suite run.
2. Regenerate evaluation sets for all five ranks at that checkpoint. At 10 000 images per rank this is ≈ 16 h on two A100s; at the full 24 995, ≈ 39 h.
3. Final comparison: real vs frozen vs five ranks on the complete suite.

Open questions for discussion.

- *Peak learning rate not scaled with rank.* All ranks share the same cosine schedule peaking at 2×10^{-4} , and `lora_alpha=rank` makes the LoRA scale factor $\alpha/r = 1$ everywhere. A rank-128 adapter therefore has $16\times$ the parameters of rank-8 taking steps of the same size, so its effective update to the UNet is larger. The rank axis thus mixes capacity with update magnitude. Two ways to resolve it: scale the peak LR with rank (e.g. $2 \times 10^{-4} \sqrt{8/r}$), at the cost of LR no longer being constant across the sweep; or stop at $\approx 3\,000\text{--}6\,000$ steps, where the sweep shows every rank is at its best and before the high-LR phase does damage. The latter keeps one more variable fixed and is cheaper.
- *Where the interesting variation lies.* Given saturation below rank 16, ranks 1, 2 and 4 would

likely be more informative than rank 256.

- *Evaluation set size.* 10 000 is the standard FID operating point and would cut generation time by $\approx 60\%$ relative to 24 995.
- *Error bars.* FID/KID are currently point estimates. KID subset resampling would give variance, which matters because several rank differences may not be significant at $n = 500$.

Appendix: infrastructure notes

- **Only renyi can run the main environment.** Its torch is 2.11.0+cu130 (needs driver ≥ 580). Drivers: renyi 580.95.05 ✓, erdos 560.35.05 ×, neumann 545.23.08 ×. On the wrong node the job silently falls back to CPU. A second environment with torch 2.4.1+cu121 unlocks the V100 nodes.
- **Throughput** (768^2 , 30 steps): A100 0.44 img/s; V100 0.07–0.09 img/s sustained — the A100 is $\approx 5\times$ faster.
- **Hardware equivalence verified.** The same seed on A100 and V100 gives the same image to within floating-point noise (mean abs. difference 0.4–0.9 / 255, versus 54 / 255 for two genuinely different samples), so mixed-hardware generation does not compromise the comparison.
- **Approximate GPU budget so far:** captioning 3.9 h; frozen baseline 13.9 h; training 46.6 GPU-h; checkpoint sweep generation 16.1 h; evaluation 1.3 h. Total ≈ 82 GPU-hours.